EXPERIMENT:7 Structures and Union

Q1.Write a C program that uses functions to perform the following operations:

a) Reading a complex number.

b) Writing a complex number.

c) Addition and substraction of two complex numbers

NOTE:represent complex number using a structure.

Algorithm:

Step 1: Start Step 2: Define a structure complex with fields:

- real
- imag

Step 3: Create functions:

- readComplex() → reads a complex number
- *omplex() → displays a complex number
- addComplex() → returns addition
- subComplex() → returns subtraction

Step 4: In main()

- Read two complex numbers c1 and c2

Compute:

- sum = addComplex(c1, c2)
- diff = subComplex(c1, c2)

Display both

Step 5: Stop

Code:

```
#include <stdio.h>

// structure for complex number
struct complex {
    float real;
    float imag;
};
```

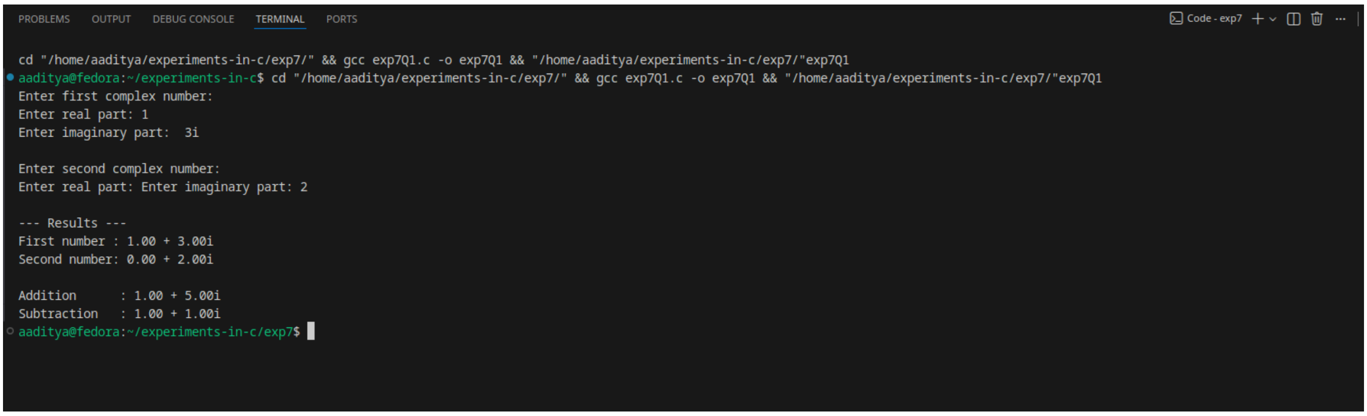
```c
// function to read a complex number
struct complex readComplex() {
    struct complex c;
    printf("Enter real part: ");
    scanf("%f", &c.real);
    printf("Enter imaginary part: ");
    scanf("%f", &c.imag);
    return c;
}

// function to write a complex number
void writeComplex(struct complex c) {
    if (c.imag >= 0)
        printf("%.2f + %.2fi\n", c.real, c.imag);
    else
        printf("%.2f - %.2fi\n", c.real, -c.imag);
}

// function for addition
struct complex addComplex(struct complex c1, struct complex c2) {
    struct complex result;
    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;
    return result;
}

// function for subtraction
struct complex subComplex(struct complex c1, struct complex c2) {
    struct complex result;
    result.real = c1.real - c2.real;
    result.imag = c1.imag - c2.imag;
    return result;
}

int main() {
    struct complex c1, c2, sum, diff;

    printf("Enter first complex number:\n");
    c1 = readComplex();

    printf("\nEnter second complex number:\n");
    c2 = readComplex();

    sum  = addComplex(c1, c2);
    diff = subComplex(c1, c2);

    printf("\n--- Results ---\n");
    printf("First number : ");
    writeComplex(c1);

    printf("Second number: ");
    writeComplex(c2);
```

```
    printf("\nAddition      : ");
    writeComplex(sum);

    printf("Subtraction   : ");
    writeComplex(diff);

    return 0;
}
```

Flowchart:

```
               ┌──────────────┐
               │    Start     │
               └───────┬──────┘
                       │
                       ▼
               ┌─────────────────────┐
               │ Read c1 (readComplex)│
               └────────┬────────────┘
                        │
                        ▼
               ┌─────────────────────┐
               │ Read c2 (readComplex)│
               └────────┬────────────┘
                        │
                        ▼
               ┌─────────────────────┐
               │ sum = addComplex     │
               └────────┬────────────┘
                        │
                        ▼
               ┌─────────────────────┐
               │ diff = subComplex    │
               └────────┬────────────┘
                        │
                        ▼
               ┌──────────────────────┐
               │ Write results         │
               └────────┬─────────────┘
                        │
                        ▼
               ┌───────────────────┐
               │       Stop         │
               └───────────────────┘
```

Output:



Q2.Write Aa C program to compute monthly pay of 100 employees using each employees name,basic pay.The DA is computed as 52% of the basic pay.Gross Salary(basic pay+DA).Print the employees name and gross salary.

Algorithm:

Step 1: Start Step 2: Declare variables:

name (string)

basic pay (float)

DA (float)

gross salary (float)

Step 3: Loop from 1 to 100 employees

3.1 Read employee name 3.2 Read basic pay 3.3 Compute DA = 0.52 × basic pay 3.4 Compute gross salary = basic pay + DA 3.5 Print employee name and gross salary

Step 4: End loop Step 5: Stop

Code:

```
#include <stdio.h>

int main() {
    char name[50];
    float basic, da, gross;

    printf("Enter details for 100 employees:\n\n");

    for (int i = 1; i <= 100; i++) {
        printf("Employee %d:\n", i);

        printf("Enter name: ");
        scanf("%s", name);      // reads name without spaces

        printf("Enter basic pay: ");
        scanf("%f", &basic);
```

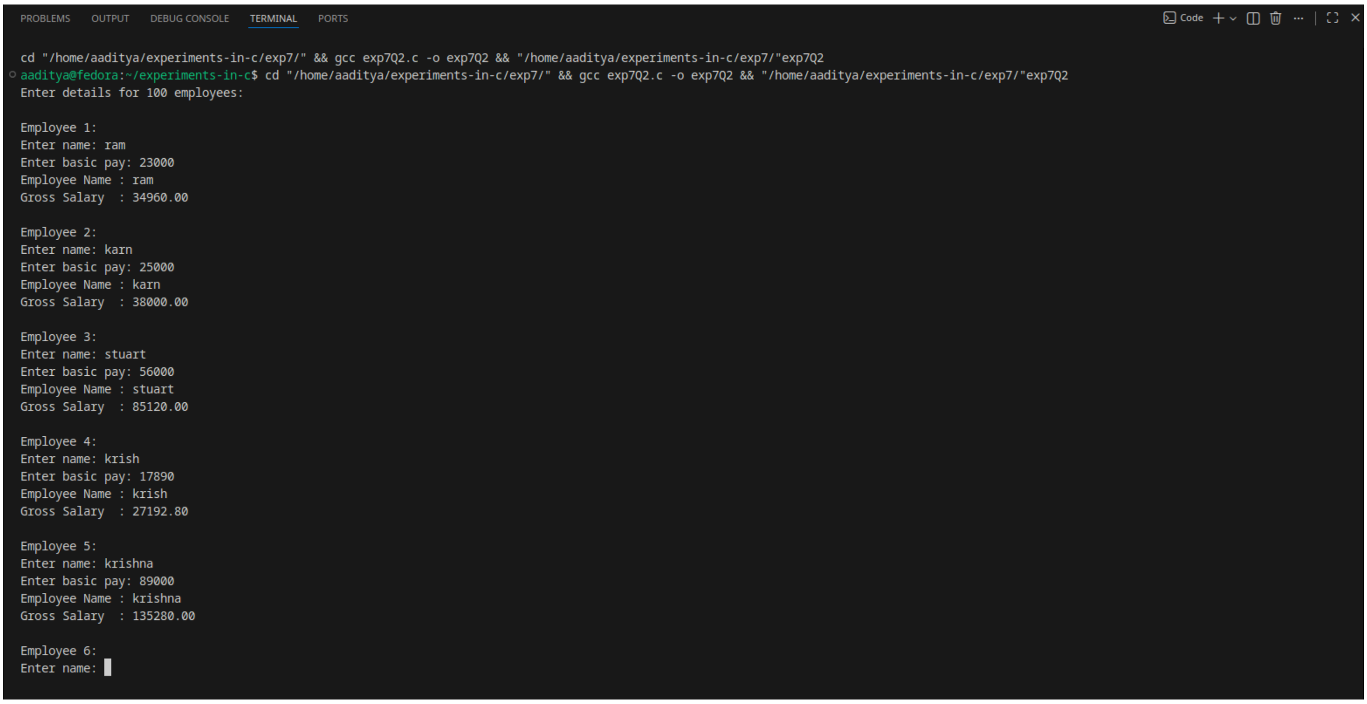
```
        da = 0.52 * basic;          // DA = 52% of basic
        gross = basic + da;         // Gross = basic + DA

        printf("Employee Name : %s\n", name);
        printf("Gross Salary  : %.2f\n\n", gross);
    }

    return 0;
}
```

Flowchart:

```
                ┌──────────────┐
                │    Start     │
                └───────┬──────┘
                        │
                ┌───────▼──────────┐
                │ i = 1            │
                └─────────┬────────┘
                          │
      ┌───────────────────▼─────────────────────┐
      │ Is i ≤ 100 ?                            │
      └────────────────┬────────────────────────┘
                       │No
                       ▼
               ┌──────────────┐
               │    Stop      │
               └──────────────┘

                       │Yes
                       ▼
       ┌─────────────────────────────┐
       │ Read employee name          │
       └───────────────┬─────────────┘
                       │
       ┌───────────────▼─────────────┐
       │ Read basic pay              │
       └───────────────┬─────────────┘
                       │
       ┌───────────────▼─────────────┐
       │ DA = 0.52 × basic           │
       └───────────────┬─────────────┘
                       │
       ┌───────────────▼─────────────┐
       │ Gross = basic + DA          │
       └───────────────┬─────────────┘
                       │
       ┌───────────────▼─────────────┐
       │ Print name and gross salary │
       └───────────────┬─────────────┘
                       │
```

```
                    |
             ┌──────▼──────┐
             │ i = i + 1    │
             └─────────────┘
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q2.c -o exp7Q2 && "/home/aaditya/experiments-in-c/exp7/"exp7Q2
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q2.c -o exp7Q2 && "/home/aaditya/experiments-in-c/exp7/"exp7Q2
Enter details for 100 employees:

Employee 1:
Enter name: ram
Enter basic pay: 23000
Employee Name : ram
Gross Salary  : 34960.00

Employee 2:
Enter name: karn
Enter basic pay: 25000
Employee Name : karn
Gross Salary  : 38000.00

Employee 3:
Enter name: stuart
Enter basic pay: 56000
Employee Name : stuart
Gross Salary  : 85120.00

Employee 4:
Enter name: krish
Enter basic pay: 17890
Employee Name : krish
Gross Salary  : 27192.80

Employee 5:
Enter name: krishna
Enter basic pay: 89000
Employee Name : krishna
Gross Salary  : 135280.00

Employee 6:
Enter name:
```

Q3.Create a book structure contanining book_id,title,author name and price.Write a C program to pass structure as a function argument and print the details.

Algorithm:

Step 1: Start Step 2: Define a structure Book with fields:

- book_id (int)
- title (string)
- author (string)
- price (float)

Step 3: Create a function displayBook(Book b) to print the details. Step 4: In main()

- Declare a variable of type Book
- Input book details (id, title, author, price)
- Call displayBook(b) by passing structure as argument

End program

Step 5: Stop

Code:

```
#include <stdio.h>

// structure definition
struct Book {
    int book_id;
    char title[50];
    char author[50];
    float price;
};

// function to display book details (structure as argument)
void displayBook(struct Book b) {
    printf("\n--- Book Details ---\n");
    printf("Book ID   : %d\n", b.book_id);
    printf("Title     : %s\n", b.title);
    printf("Author    : %s\n", b.author);
    printf("Price     : %.2f\n", b.price);
}

int main() {
    struct Book b;

    printf("Enter Book ID: ");
    scanf("%d", &b.book_id);

    printf("Enter Book Title: ");
    scanf("%s", b.title);

    printf("Enter Author Name: ");
    scanf("%s", b.author);

    printf("Enter Price: ");
    scanf("%f", &b.price);

    // pass structure to function
    displayBook(b);

    return 0;
}
```

Flowchart:

```
                    ┌──────────────┐
                    │    Start     │
                    └───────┬──────┘
                            │
                            ▼
                ┌───────────────────────┐
                │ Input book_id         │
```

```
 └──────────────────────────┘
              │
              │
 ┌────────────▼─────────────┐
 │ Input title              │
 └──────────────────────────┘
              │
              │
 ┌────────────▼─────────────┐
 │ Input author             │
 └──────────────────────────┘
              │
              │
 ┌────────────▼─────────────┐
 │ Input price              │
 └──────────────────────────┘
              │
              │
 ┌────────────▼──────────────┐
 │ Call displayBook(b)       │
 └───────────────────────────┘
              │
              │
     ┌────────▼────────┐
     │      Stop       │
     └─────────────────┘
```

Output:


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - exp7

```
cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q3.c -o exp7Q3 && "/home/aaditya/experiments-in-c/exp7/"exp7Q3
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q3.c -o exp7Q3 && "/home/aaditya/experiments-in-c/exp7/"exp7Q3
Enter Book ID: 1098
Enter Book Title: harry potter
Enter Author Name: Enter Price: jk rowling

--- Book Details ---
Book ID   : 1098
Title     : harry
Author    : potter
Price     : 0.00
aaditya@fedora:~/experiments-in-c/exp7$
```

Q4.Create a Union containing 6 strings : name,home_address,hostel_address,city ,state and zip.Write a C program to display your present address.

Algorithm:

Step 1: Start Step 2: Define a union Address with members:

name

home_address

hostel_address

city

state

zip

Step 3: Declare a variable of union type (A). Step 4: Copy a string into A.name using strcpy()

Print the value

Step 5: Copy a string into A.home_address

Print the value

Step 6: Copy a string into A.hostel_address

Print

Step 7: Copy a string into A.city

Print

Step 8: Copy a string into A.state

Print

Step 9: Copy a string into A.zip

Print

Step 10: Stop

Code:

```
#include <stdio.h>
#include <string.h>

union Address {
    char name[50];
    char home_address[100];
    char hostel_address[100];
    char city[50];
    char state[50];
    char zip[10];
};

int main() {
    union Address A;

    printf("---- Present Address ----\n");

    strcpy(A.name, "Aaditya Mishra");
    printf("Name          : %s\n", A.name);

    strcpy(A.home_address, "123, MG Road");
    printf("Home Address  : %s\n", A.home_address);

    strcpy(A.hostel_address, "Hostel Block B");
    printf("Hostel Address: %s\n", A.hostel_address);

    strcpy(A.city, "Ranchi");
    printf("City          : %s\n", A.city);
```

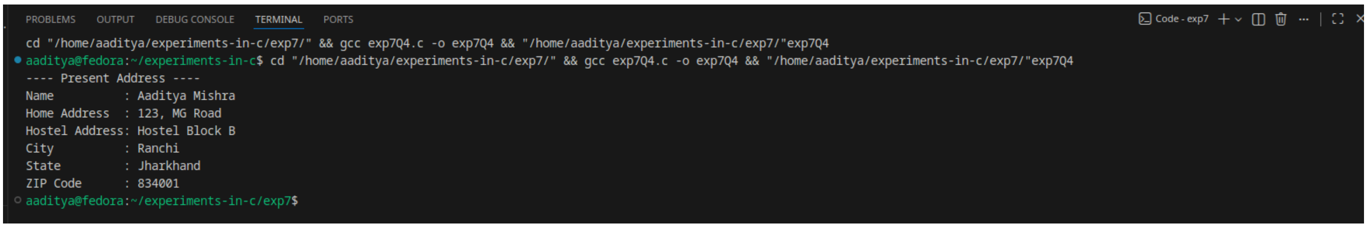
```
    strcpy(A.state, "Jharkhand");
    printf("State         : %s\n", A.state);

    strcpy(A.zip, "834001");
    printf("ZIP Code      : %s\n", A.zip);

    return 0;
}
```

Flowchart:

```
                ┌──────────────┐
                │    Start     │
                └───────┬──────┘
                        │
                        │
        ┌───────────────▼──────────────────┐
        │ Define union Address with fields │
        │ name, home_address, etc.         │
        └───────────────┬──────────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Declare A (union var)    │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Copy string to A.name    │
            │ Print A.name             │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Copy to A.home_address   │
            │ Print home_address       │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Copy to A.hostel_addr    │
            │ Print hostel_address     │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Copy to A.city           │
            │ Print city               │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
            │ Copy to A.state          │
            │ Print state              │
            └───────────┬──────────────┘
                        │
                        │
            ┌───────────▼──────────────┐
```

```
        │  Copy to A.zip              │
        │  Print ZIP code             │
        └─────────────┬───────────────┘
                      │
                      │
              ┌───────▼───────┐
              │     Stop      │
              └───────────────┘
```

Output:



```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q4.c -o exp7Q4 && "/home/aaditya/experiments-in-c/exp7/"exp7Q4
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp7/" && gcc exp7Q4.c -o exp7Q4 && "/home/aaditya/experiments-in-c/exp7/"exp7Q4
---- Present Address ----
Name          : Aaditya Mishra
Home Address  : 123, MG Road
Hostel Address: Hostel Block B
City          : Ranchi
State         : Jharkhand
ZIP Code      : 834001
aaditya@fedora:~/experiments-in-c/exp7$
```